

Student Academic Performance Prediction Using Machine Learning and Deep Learning Techniques

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Abstract

Student performance prediction has become an important application of educational data mining (EDM) because it enables educational institutions to identify students who may require academic support before poor outcomes occur. Early prediction allows instructors and institutions to implement interventions that improve retention, academic success, and overall student achievement. This study investigates the effectiveness of traditional machine learning and deep learning approaches for predicting student academic performance using the UCI Student Performance dataset.

The dataset contains 395 student records with demographic, family, social, and academic variables. A binary classification problem was formulated by converting the final grade variable into a pass/fail target. Data preprocessing included feature engineering, categorical encoding, feature scaling, and train-test splitting. Seven predictive models were evaluated, including logistic regression, logistic regression with modified regularization, random forest, tuned random forest, sequential neural networks, sequential neural networks with dropout, and a functional API neural network.

Model performance was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Results demonstrated that traditional machine learning models outperformed deep learning approaches on this relatively small tabular dataset. Logistic regression achieved the highest ROC-AUC score of 0.948, while the tuned random forest achieved the highest F1-score of 0.902. Deep learning models produced competitive results but did not surpass traditional methods.

The findings suggest that simpler models can provide excellent predictive performance when working with structured educational datasets and limited sample sizes. The study highlights the

importance of selecting algorithms based on dataset characteristics rather than model complexity alone.

Keywords: Educational Data Mining, Student Performance Prediction, Machine Learning, Deep Learning, Logistic Regression, Random Forest, and Neural Networks.

1. Introduction

The increasing availability of educational data has transformed the way institutions evaluate student performance and make academic decisions. Educational institutions collect large amounts of information related to student demographics, attendance, study habits, family background, and academic achievement. Advances in machine learning have created opportunities to leverage this information to improve educational outcomes through predictive analytics.

Student performance prediction is one of the most widely studied applications of educational data mining (EDM). The ability to identify students who may be at risk of academic failure before final assessments occur allows educators to provide targeted support and intervention. Such interventions may include tutoring programs, academic counseling, mentoring, or personalized learning strategies. Consequently, predictive models have the potential to improve both individual student outcomes and institutional performance.

Traditional statistical methods have historically been used to analyze educational datasets. However, machine learning techniques have gained popularity because of their ability to discover patterns and relationships that may not be evident through conventional analysis. Algorithms such as logistic regression, decision trees, support vector machines, and random forests have been successfully applied to educational prediction problems. These models can learn complex relationships among variables and generate predictions with high accuracy.

In recent years, deep learning has emerged as one of the most influential areas of artificial intelligence. Neural networks have demonstrated remarkable success in image recognition, speech processing, natural language understanding, and many other domains. Because of these successes, researchers have increasingly explored the use of deep learning techniques for educational prediction tasks. Neural networks offer the ability to model nonlinear relationships and interactions among features that may not be captured by simpler algorithms.

Despite the growing interest in deep learning, there remains debate regarding its effectiveness for educational datasets. Many educational datasets are relatively small and consist primarily of structured tabular data. Under such conditions, traditional machine learning methods often remain competitive and may even outperform more complex neural network architectures. Understanding the circumstances under which each approach performs best is therefore an important research question.

This study investigates student performance prediction using the UCI Student Performance dataset. The dataset contains information about secondary school students, including demographic characteristics, family background, social activities, and academic records. The primary objective is to determine whether machine learning and deep learning models can accurately predict whether a student will pass or fail based on available features.

The study evaluates seven predictive models representing different levels of complexity. These include Logistic Regression, Logistic Regression with modified regularization, Random Forest, Tuned Random Forest, Sequential Neural Networks, Sequential Neural Networks with Dropout, and a Functional API Neural Network. Performance is assessed using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

The specific objectives of this study are:

1. To preprocess and prepare educational data for predictive modeling.
2. To implement and evaluate multiple machine learning and deep learning algorithms.
3. To compare model performance using standardized evaluation metrics.
4. To identify the most effective approach for predicting student academic performance.
5. To analyze the strengths and limitations of traditional machine learning and deep learning methods within an educational context.

The findings of this study contribute to the growing body of research in Educational Data Mining by providing a direct comparison between traditional machine learning and deep learning approaches using a common dataset and evaluation framework.

2. Literature Review

Educational Data Mining (EDM) is an interdisciplinary field that combines data science, statistics, machine learning, and educational research to extract meaningful insights from educational data. The primary objective of EDM is to improve learning outcomes by understanding student behavior, identifying performance trends, and supporting data-driven decision-making.

One of the foundational studies in student performance prediction was conducted by Cortez and Silva, who introduced the Student Performance dataset used in this project. Their research demonstrated that academic performance could be predicted using demographic, social, and academic variables. The study highlighted the importance of educational data mining in identifying factors that influence student success and provided a benchmark dataset that continues to be widely used in predictive modeling research.

Research by Baker and Yacef emphasized the role of educational data mining in improving educational outcomes. They argued that data-driven approaches can help institutions better

understand student learning patterns and support evidence-based interventions. Their work helped establish EDM as a recognized research discipline and encouraged broader adoption of predictive analytics in education.

Traditional machine learning methods have been widely applied to student performance prediction. Logistic Regression remains one of the most commonly used algorithms because of its simplicity, interpretability, and effectiveness. Logistic Regression models the probability of class membership and is particularly suitable for binary classification tasks such as pass/fail prediction. Several studies have demonstrated that Logistic Regression provides strong baseline performance while offering transparency in decision-making.

Random Forest has emerged as another highly effective approach for educational prediction tasks. Introduced by Breiman, Random Forest combines multiple decision trees to improve predictive accuracy and reduce overfitting. The ensemble learning approach allows Random Forest to capture nonlinear relationships among variables while maintaining strong generalization performance. Numerous studies report that Random Forest consistently achieves high accuracy on educational datasets and often outperforms individual decision trees.

Deep learning methods have gained attention because of their ability to learn complex feature representations automatically. Artificial Neural Networks (ANNs) consist of interconnected layers of neurons that transform input data through multiple nonlinear operations. Researchers have applied neural networks to student performance prediction, dropout detection, and learning analytics. Deep learning models are particularly attractive because they can model intricate relationships among variables without requiring extensive manual feature engineering.

However, several studies suggest that deep learning does not always outperform traditional machine learning methods. Educational datasets are often characterized by limited sample sizes and structured tabular features. Under such conditions, neural networks may struggle to generalize effectively and may be more susceptible to overfitting. Consequently, simpler algorithms such as Logistic Regression and Random Forest frequently remain competitive.

Comparative studies evaluating machine learning and deep learning approaches have produced mixed results. Some researchers report modest improvements from neural networks, while others find that traditional methods achieve comparable or superior performance. These findings indicate that model effectiveness depends heavily on dataset characteristics, including size, feature complexity, and data quality.

Another important area of research concerns identifying the factors most strongly associated with student success. Previous studies consistently identify prior academic achievement, study habits, attendance, family support, parental education, and socioeconomic status as significant predictors of academic performance. Understanding these relationships is valuable because it enables institutions to design targeted interventions that address specific student needs.

Although substantial progress has been made in student performance prediction, several challenges remain. Many studies rely on datasets collected from specific institutions or regions,

limiting the generalizability of findings. Additionally, there is a need for more comprehensive comparisons between traditional machine learning and deep learning approaches using consistent evaluation frameworks.

This study addresses these challenges by evaluating multiple machine learning and deep learning models on the same dataset using identical preprocessing procedures and evaluation metrics. The comparison provides insight into the relative strengths and weaknesses of each approach and contributes to the ongoing discussion regarding model selection for educational prediction tasks.

3. Dataset Description

3.1 Dataset Overview

This study utilizes the Student Performance Dataset obtained from the UCI Machine Learning Repository. The dataset was originally collected by Cortez and Silva and has become one of the most widely used educational datasets for predictive modeling research.

The dataset contains information about secondary school students and includes variables describing demographic characteristics, family background, educational environment, social behavior, and academic performance. These features provide valuable insights into factors that may influence student success and academic achievement.

Dataset Characteristics

Attribute	Value
Dataset Name	Student Performance Dataset
Source	UCI Machine Learning Repository
Total Records	395
Total Features	33
Target Variable	Pass/Fail Classification
Problem Type	Binary Classification

The original dataset contains the final grade variable (G3), which represents student academic performance. To formulate a classification problem suitable for machine learning algorithms, G3 was transformed into a binary target variable representing whether a student passed or failed.

Students with a final grade greater than or equal to 10 were classified as passing students, while students with grades below 10 were classified as at-risk students. This transformation created a binary classification task focused on identifying students who may require academic intervention.

3.2 Feature Categories

The dataset contains multiple categories of variables that describe different aspects of student life and academic experience.

Demographic Features

Demographic variables include:

- Age
- Gender
- Residential location
- Family size
- Parent cohabitation status

These variables provide information about the personal characteristics and living conditions of students.

Family Background Features

Family-related variables include:

- Mother's education
- Father's education
- Mother's occupation
- Father's occupation
- Family support
- Family relationship quality

Previous research suggests that family background plays an important role in educational achievement.

Academic Features

Academic variables include:

- Study time
- Number of previous failures
- Educational support
- School support

- Extra paid classes

These features directly relate to learning activities and academic preparation.

Social Features

Social and behavioral variables include:

- Free time
- Social activities
- Alcohol consumption
- Internet access
- Romantic relationships

These factors may indirectly influence academic performance by affecting study habits and student engagement.

Grade Variables

The dataset includes:

- G1 (First Period Grade)
- G2 (Second Period Grade)
- G3 (Final Grade)

G3 was used to create the binary target variable for this project.

4. Methodology

4.1 Research Design

This study follows a supervised machine learning framework for binary classification. The objective is to predict whether a student will pass or fail based on demographic, family, social, and academic attributes.

The workflow consists of the following stages:

1. Data collection and inspection
2. Data preprocessing
3. Feature engineering
4. Model development

5. Model evaluation
6. Comparative analysis

The same dataset and preprocessing pipeline were used for all experiments to ensure a fair comparison among models.

4.2 Data Preprocessing

Data preprocessing is an essential step in machine learning because model performance depends heavily on data quality and representation.

Missing Value Inspection

The dataset was inspected for missing values before model training. Data quality checks confirmed that the dataset contained minimal missing information and required limited cleaning.

Target Variable Engineering

The original final grade variable (G3) was converted into a binary classification target.

The transformation was defined as:

- Pass = $G3 \geq 10$
- Fail = $G3 < 10$

This conversion simplified the prediction task and aligned the project with educational intervention objectives.

Categorical Feature Encoding

Many features within the dataset are categorical and cannot be processed directly by machine learning algorithms.

Examples include:

- Gender
- School
- Address type
- Family support
- Internet access

These variables were converted into numerical representations using One-Hot Encoding.

One-Hot Encoding creates binary indicator variables that preserve categorical information without introducing artificial ordering.

Feature Scaling

Numerical variables were standardized using StandardScaler.

Feature scaling ensures that variables measured on different scales contribute equally during model training. Standardization transforms variables to have:

- Mean = 0
- Standard Deviation = 1

This step is particularly important for Logistic Regression and Neural Networks.

Train-Test Split

The dataset was divided into training and testing subsets using an 80/20 ratio.

- Training Set: 80%
- Testing Set: 20%

Stratified sampling was used to preserve class distribution across both subsets.

The testing dataset remained unseen during training and was used exclusively for final model evaluation.

4.3 Evaluation Metrics

To ensure a comprehensive assessment of model performance, multiple evaluation metrics were used.

Accuracy

Accuracy measures the proportion of correctly classified observations.

Accuracy = $\text{Correct Predictions} / \text{Total Predictions}$

Although widely used, accuracy alone may be insufficient when class distributions are imbalanced.

Precision

Precision measures the proportion of positive predictions that are correct.

Precision is particularly useful when false positives are costly.

Recall

Recall measures the proportion of actual positive cases correctly identified.

In educational applications, recall is important because failing to identify at-risk students may prevent timely intervention.

F1-Score

The F1-score combines precision and recall into a single metric.

This measure is useful when balancing false positives and false negatives.

ROC-AUC

The Receiver Operating Characteristic Area Under the Curve (ROC-AUC) evaluates a model's ability to distinguish between classes across different classification thresholds.

A ROC-AUC value above 0.90 is generally considered excellent.

5. Experimental Setup

5.1 Experimental Framework

The experimental framework was designed to compare traditional machine learning methods with deep learning approaches using identical datasets and evaluation criteria.

Seven experiments were conducted.

Each experiment represents a different modeling strategy with varying levels of complexity.

5.2 Experiment 1: Logistic Regression Baseline

Logistic Regression served as the baseline model.

The model was selected because:

- It is widely used in binary classification.
- It provides interpretable results.
- It serves as a strong benchmark for comparison.

The baseline model was trained using default parameters.

5.3 Experiment 2: Logistic Regression with Modified Regularization

The second experiment modified the regularization strength parameter ($C = 0.5$).

The objective was to investigate the impact of stronger regularization on predictive performance and generalization.

5.4 Experiment 3: Random Forest Default

The third experiment utilized a Random Forest classifier with default hyperparameters.

Random Forest combines multiple decision trees through ensemble learning to improve predictive performance and reduce overfitting.

5.5 Experiment 4: Tuned Random Forest

The fourth experiment applied hyperparameter tuning to optimize Random Forest performance.

Parameters such as:

- Number of trees
- Maximum tree depth
- Minimum samples per split

were adjusted to improve model generalization.

5.6 Experiment 5: Deep Sequential Neural Network

The first deep learning experiment used a Sequential Neural Network architecture implemented using TensorFlow and Keras.

The architecture consisted of:

- Input layer
- Hidden dense layers
- Output layer with sigmoid activation

Binary cross-entropy was used as the loss function.

5.7 Experiment 6: Deep Sequential Network

A deeper neural network architecture was evaluated to determine whether increased representational capacity could improve predictive performance.

Additional hidden layers were introduced to capture more complex relationships among variables.

5.8 Experiment 7: Sequential Neural Network with Dropout

The seventh experiment incorporated Dropout regularization.

Dropout randomly deactivates neurons during training, reducing overfitting and improving generalization.

The experiment investigated whether regularization could improve neural network performance on a relatively small dataset.

5.9 Experiment 8: Functional API Neural Network

The final experiment utilized TensorFlow's Functional API.

The Functional API provides greater architectural flexibility than the Sequential framework and supports more complex model structures.

This experiment evaluated whether architectural flexibility would improve predictive performance compared with simpler neural network configurations.

6. Results

6.1 Experimental Results

The performance of all models was evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC. These metrics provide a comprehensive assessment of classification performance and allow meaningful comparisons across different modeling approaches.

Table 1: Model Performance Comparison

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score	ROC-AUC
Logistic Regression Baseline	87.34	95.74	84.91	0.900	0.948
Logistic Regression (C=0.5)	86.08	94.34	84.91	0.889	0.944
Random Forest Default	86.08	92.73	86.44	0.891	0.929
Random Forest Tuned	87.34	92.86	87.93	0.902	0.922
Sequential Deep	84.81	92.59	84.75	0.882	0.927

Sequential Dropout	83.54	91.07	83.05	0.871	0.934
Functional API	82.28	89.29	83.05	0.860	0.924

The results demonstrate that all models achieved strong predictive performance, with every model exceeding 82% accuracy and 0.92 ROC-AUC. This indicates that the dataset contains meaningful patterns that can be successfully leveraged for student performance prediction.

Among all models, Logistic Regression Baseline achieved the highest ROC-AUC score of 0.948, while the Tuned Random Forest achieved the highest F1-score of 0.902. These results suggest that traditional machine learning methods remain highly competitive for structured educational datasets.

7. Discussion

The primary objective of this study was to compare traditional machine learning and deep learning approaches for student academic performance prediction. The experimental results provide several important insights regarding model effectiveness, generalization, and suitability for educational datasets.

A notable finding is that traditional machine learning methods outperformed deep learning models. Logistic Regression and Tuned Random Forest consistently achieved the strongest overall performance. This outcome may initially appear surprising given the success of deep learning in many domains. However, the result aligns with existing research indicating that deep learning does not always outperform simpler methods when datasets are relatively small and highly structured.

Logistic Regression achieved the highest ROC-AUC score of 0.948 and matched the highest overall accuracy of 87.34%. The model demonstrated excellent discrimination between passing and failing students. Its strong performance suggests that many of the relationships within the dataset can be effectively captured using linear decision boundaries. Furthermore, Logistic Regression benefits from low computational requirements and high interpretability, making it particularly attractive for educational applications where transparency is important.

The Tuned Random Forest achieved the highest F1-score of 0.902. This indicates a strong balance between precision and recall and highlights the effectiveness of hyperparameter optimization. By combining multiple decision trees, the Random Forest model was able to

capture nonlinear interactions among variables while maintaining strong generalization performance.

The neural network models achieved respectable performance but failed to surpass the best traditional machine learning models. The Deep Sequential Network achieved an accuracy of 84.81%, while the Functional API model achieved 82.28%. These results suggest that the dataset size was insufficient to fully exploit the representational power of deep learning architectures.

Another important observation is that increasing model complexity did not necessarily improve performance. The Functional API model, despite its architectural flexibility, produced the lowest overall accuracy among all evaluated models. This reinforces the principle that model complexity should be matched to dataset characteristics rather than assumed to provide automatic improvements.

Overall, the findings demonstrate that carefully tuned traditional machine learning models remain highly effective for educational prediction tasks involving structured tabular data.

8. Learning Curve Analysis

Learning curves provide valuable insight into model training behavior and generalization performance.

Logistic Regression

The learning curves for Logistic Regression converged smoothly, with training and validation performance remaining relatively close throughout training. This behavior suggests that the model achieved a favorable balance between bias and variance. The absence of significant divergence indicates strong generalization and limited overfitting.

Random Forest

The Random Forest model exhibited higher training performance than validation performance, suggesting moderate variance. Hyperparameter tuning successfully reduced the gap between training and validation performance, improving generalization and contributing to the model's strong F1-score.

Deep Learning Models

The neural network learning curves showed steady improvements in training accuracy during early epochs. Validation accuracy also improved initially but eventually plateaued while training accuracy continued increasing. This pattern indicates mild overfitting. The introduction of Dropout regularization reduced this divergence and improved generalization, although overall predictive performance remained slightly lower than traditional machine learning models.

9. Confusion Matrix Analysis

Confusion matrices provide insight into the types of classification errors made by predictive models.

Logistic Regression

The confusion matrix demonstrated that most students were correctly classified. The relatively small number of false negatives indicates that the model effectively identified students who may be academically at risk. This is particularly important because false negatives represent students who require support but may not be identified for intervention.

Random Forest

The Tuned Random Forest produced a balanced distribution of false positives and false negatives. This balance contributed to its strong F1-score and indicates that the model effectively managed the trade-off between precision and recall.

Deep Learning Models

The neural network models generated a greater number of classification errors than the top-performing traditional machine learning models. While their performance remained competitive, the additional complexity did not result in fewer misclassifications.

10. ROC Curve Analysis

Receiver Operating Characteristic (ROC) analysis evaluates a model's ability to distinguish between classes across different classification thresholds.

All evaluated models achieved ROC-AUC values greater than 0.92, indicating excellent discriminative capability.

Table 2: ROC-AUC Comparison

Model	ROC-AUC
Logistic Regression Baseline	0.948
Logistic Regression C=0.5	0.944
Sequential Dropout	0.934
Random Forest Default	0.929
Sequential Deep	0.927
Functional API	0.924
Random Forest Tuned	0.922

The logistic regression model achieved the highest ROC-AUC score, demonstrating superior ability to rank passing and failing students across classification thresholds.

11. Bias-Variance Analysis

The results clearly illustrate the classical bias-variance trade-off.

- Logistic regression exhibited relatively high bias but low variance, resulting in strong generalization performance.
- Random Forest reduced bias by modeling nonlinear patterns but introduced additional variance.
- Deep neural networks further reduced bias through increased representational capacity but experienced higher variance due to the limited dataset size.
- Dropout regularization successfully reduced variance by preventing excessive reliance on individual neurons.

The strong performance of logistic regression suggests that the relationships within the dataset are sufficiently simple to be captured by a relatively low-complexity model.

12. Limitations

Several limitations should be considered when interpreting the findings.

1. The dataset contains only 395 observations, limiting the amount of information available for training complex models.
2. The data were collected within a specific educational context and may not generalize to other institutions or countries.
3. Several variables rely on self-reported information and may contain measurement bias.
4. The binary pass/fail transformation simplifies academic performance and may conceal important distinctions among students.
5. Additional factors such as instructional quality, curriculum design, and socioeconomic conditions were not fully represented.

These limitations should be considered when applying the findings to broader educational settings.

13. Future Work

Several opportunities exist for extending this research.

Future studies may consider:

- Larger and more diverse educational datasets.
- Advanced ensemble techniques such as XGBoost and LightGBM.
- Explainable Artificial Intelligence methods, including SHAP and LIME.

- Automated feature selection techniques.
- Longitudinal educational datasets containing multiple semesters of student records.
- Transfer learning approaches and automated hyperparameter optimization.

Such improvements may further enhance predictive performance and practical applicability.

14. Conclusion

This study investigated the use of machine learning and deep learning techniques for predicting student academic performance using the UCI Student Performance dataset. Seven predictive models were implemented and evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

The results demonstrated that traditional machine learning approaches outperformed deep learning models. Logistic Regression achieved the highest ROC-AUC score of 0.948, while the Tuned Random Forest achieved the highest F1-score of 0.902. Deep learning models achieved competitive performance but did not provide sufficient gains to justify their increased complexity.

The findings highlight the importance of selecting predictive models based on dataset characteristics rather than algorithm sophistication alone. For relatively small and structured educational datasets, traditional machine learning methods remain highly effective and practical solutions.

Overall, the project demonstrates the potential of predictive analytics to support educational decision-making and identify students who may benefit from academic intervention. By leveraging data-driven approaches, educational institutions can improve student outcomes and allocate support resources more effectively.

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